Name:P.Pranay H.No:2303A51829 Batch:26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s) Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju		
CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week10 – Wednesday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
Assignment Number:19.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*Expected Time to complete*
1	**Lab 19 – Code Translation: Converting Between Programming Languages** **Lab Objectives:** • Understand how AI tools can assist in translating code between	Week10 - Wednesday

different programming languages.

- Learn to verify correctness and functionality after translation.
- Explore syntactic and semantic differences between languages (e.g., Python, Java, C++).
- Practice debugging and optimizing AI-translated code.

Task 1 – Python to C++ Conversion

Task:

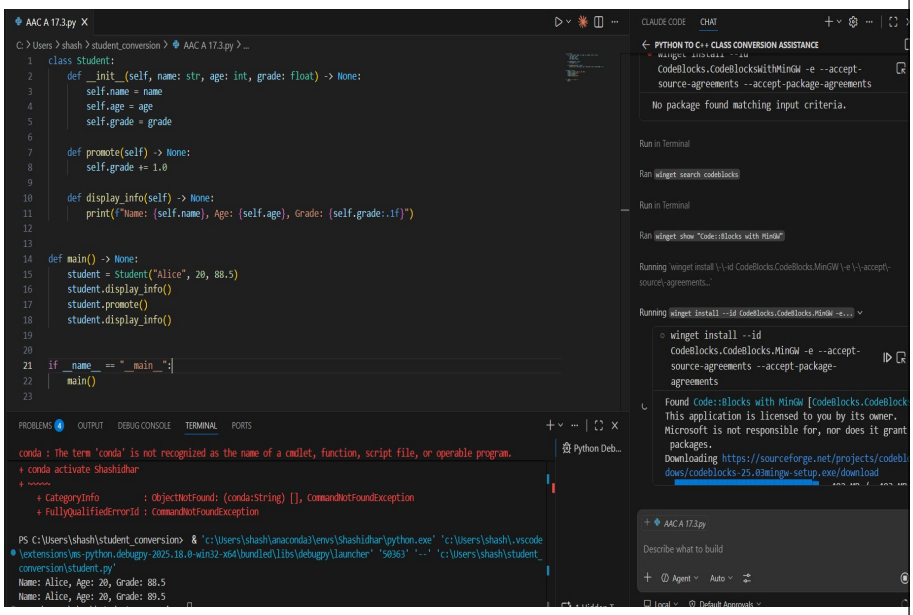
Translate a simple Python class into C++ using AI assistance.

Instructions:

- Prompt AI to convert a Python class representing a Student with attributes and methods into equivalent C++ code
- Ensure proper handling of:
  - Constructors
  - Data types
  - Access specifiers
- Compile and run both versions to verify output consistency

Expected Output:

An equivalent working C++ class translated from Python.



Task 2 – Java to Python Function Conversion

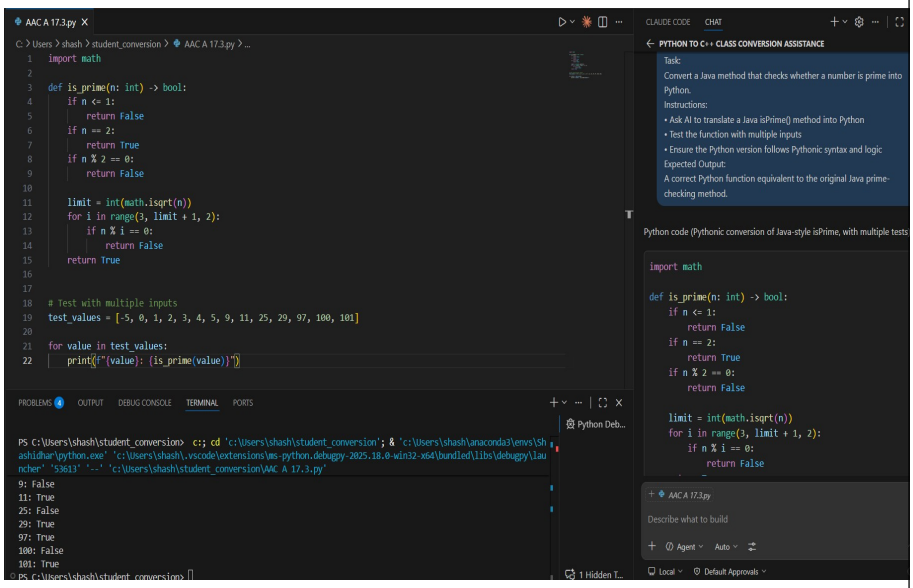
Task:

Convert a Java method that checks whether a number is prime into Python.

Instructions:

- Ask AI to translate a Java isPrime() method into Python
- Test the function with multiple inputs
- Ensure the Python version follows Pythonic syntax and logic

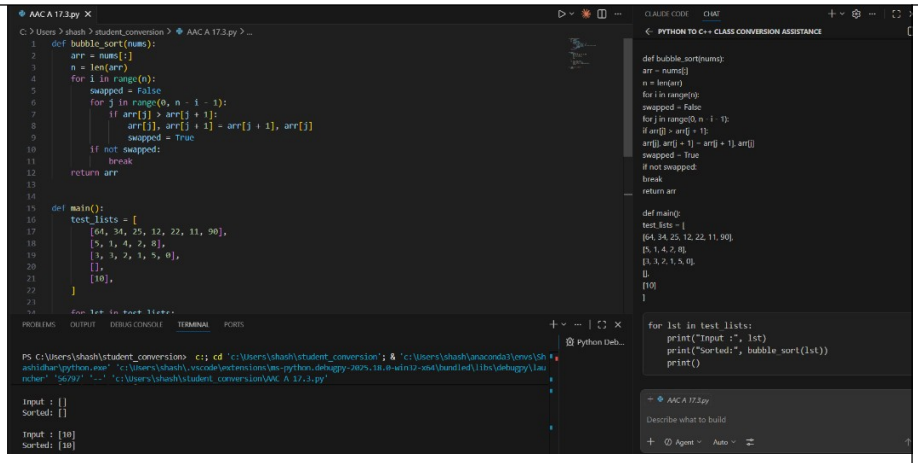
Expected Output:
A correct Python function equivalent to the original Java prime-checking method.

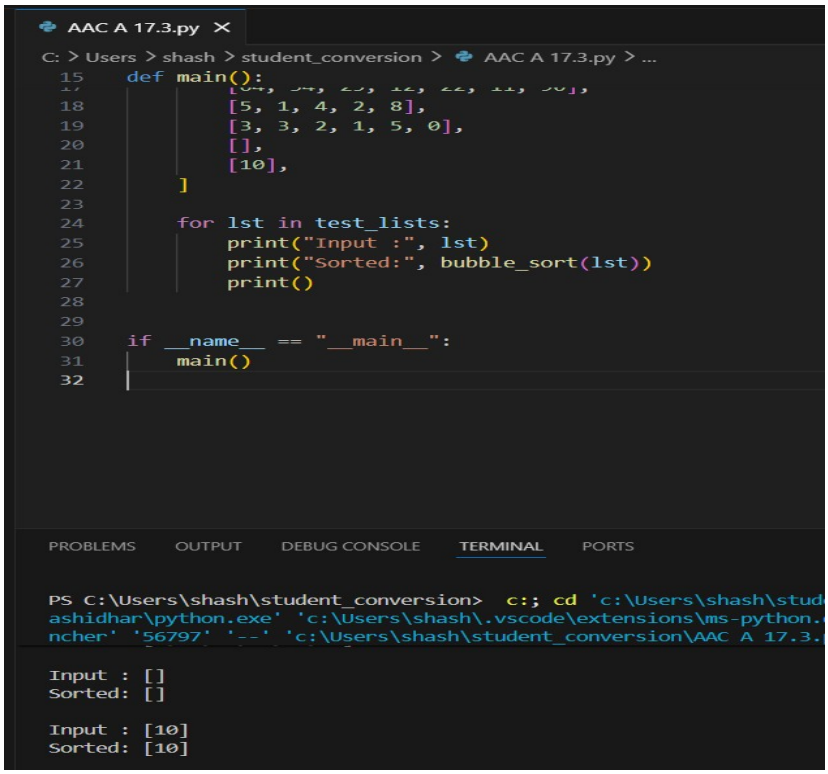


Task 3 – Pseudocode to Python Implementation
Task:
Translate a given pseudocode algorithm into Python using AI.
Instructions:

- Provide AI with pseudocode for sorting numbers (e.g., bubble sort or selection sort)
- Ask AI to convert it into executable Python code
- Validate correctness using sample input lists

Expected Output:
A working Python program that correctly implements the pseudocode logic.





Task 4 – SQL to Pandas Query

Task:

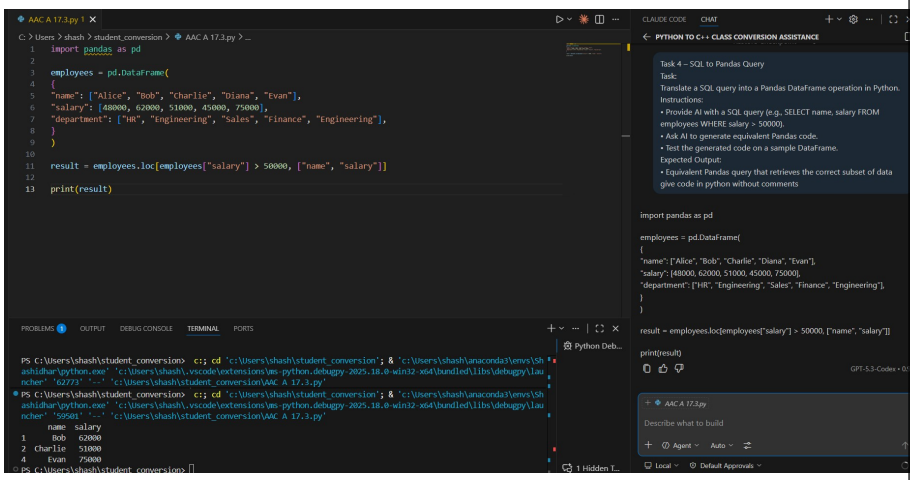
Translate a **SQL query** into a **Pandas DataFrame operation** in Python.

Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.



Task 5 – Real-Time Application: Algorithm Translation Across Languages

Scenario:

A company maintains algorithms written in different programming languages.

Examples:

- Converting a Python searching algorithm into C
- Translating a Java sorting program into JavaScript

Instructions:

- Use AI to translate a selected algorithm between two programming languages
- Execute and test both versions
- Document translation challenges such as:
  - Syntax differences
  - Library support
  - Memory management

Expected Output:

Equivalent, tested code implementations in two different programming languages.

AAC A 17.3.py

C: > Users > shash > student_conversion > AAC A 17.3.py > ...

```python
def python_binary_search(arr, target):
    left, right = 0, len(arr) - 1
    while left <= right:
        mid = left + (right - left) // 2
        if arr[mid] == target:
            return mid
        if arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1


def c_style_binary_search(arr, size, target):
    left = 0
    right = size - 1
    while left <= right:
        mid = left + (right - left) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Sh
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\lau
ncher' '49361' '--' 'c:\Users\shash\student_conversion\AAC A 17.3.py'
Search Tests
input=[64, 34, 25, 12, 22, 11, 90], java_like=[11, 12, 22, 25, 34, 64, 90], js_like=[11, 12, 22, 25, 34, 64, 90], matc
h=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
PS C:\Users\shash\student_conversion>
```

Python Deb...

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CLAUDE CODE CHAT

PYTHON TO C++ CLASS CONVERSION ASSISTANCE

Task 5 – Real-Time Application: Algorithm Translation Across Languages
Scenario:
A company maintains algorithms written in different programming languages.
Examples:
• Converting a Python searching algorithm into C
• Translating a Java sorting program into JavaScript
Instructions:
• Use AI to translate a selected algorithm between two programming languages
• Execute and test both versions
• Document translation challenges such as:
o Syntax differences
o Library support
o Memory management
Expected Output:
Equivalent, tested code implementations in two different programming languages.
give code in python without comments

```
def python_binary_search(arr, target):
left, right = 0, len(arr) - 1
while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
```

AAC A 17.3.py

Describe what to build

Agent Auto

Local Default Approvals

AAC A 17.3.py

C: > Users > shash > student_conversion > AAC A 17.3.py > ...

```python
def c_style_binary_search(arr, size, target):
    left = 0
    right = size - 1
    while left <= right:
        mid = left + (right - left) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1


def java_bubble_sort(arr):
    a = arr[:]
    n = len(a)
    for i in range(n - 1):
        for j in range(0, n - i - 1):
            if a[j] > a[j + 1]:
                temp = a[j]
                a[j] = a[j + 1]
                a[j + 1] = temp
    return a
```

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input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
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```

Python Deb...

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CLAUDE CODE CHAT

PYTHON TO C++ CLASS CONVERSION ASSISTANCE

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else:
right = mid - 1
return -1

def c_style_binary_search(arr, size, target):
left = 0
right = size - 1
while left <= right:
mid = left + (right - left) // 2
if arr[mid] == target:
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left = mid + 1
else:
right = mid - 1
return -1

def java_bubble_sort(arr):
a = arr[:]
n = len(a)
for i in range(n - 1):
```

AAC A 17.3.py

Describe what to build

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AAC A 17.3.py

C: > Users > shash > student_conversion > AAC A 17.3.py > ...

```python
def javascript_bubble_sort(arr):
    a = arr[:]
    n = len(a)
    for i in range(n):
        swapped = False
        for j in range(0, n - i - 1):
            if a[j] > a[j + 1]:
                a[j], a[j + 1] = a[j + 1], a[j]
                swapped = True
        if not swapped:
            break
    return a


def run_search_tests():
    data = [3, 7, 12, 19, 24, 31, 45, 58]
    targets = [3, 24, 58, 9]
    print("Search Tests")
    for t in targets:
        p = python_binary_search(data, t)
        c = c_style_binary_search(data, len(data), t)
        print(f"target={t}, python={p}, c_style={c}, match={p == c}")

```

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```
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h=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
```

AAC A 17.3.py

C: > Users > shash > student_conversion > AAC A 17.3.py > ...

```python

def run_sort_tests():
    test_cases = [
        [64, 34, 25, 12, 22, 11, 90],
        [5, 1, 4, 2, 8],
        [3, 3, 2, 1, 5, 0],
        [],
        [10],
    ]
    print("\nSort Tests")
    for case in test_cases:
        j = java_bubble_sort(case)
        js = javascript_bubble_sort(case)
        print(f"input={case}, java_like={j}, js_like={js}, match={j == js}")


if __name__ == "__main__":
    run_search_tests()
    run_sort_tests()

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion>  c:; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy
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input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.